

HOT WORK TOOLBOX TALK

Prevent fire and explosion during welding, cutting and grinding.

8-10 MINUTES

OBJECTIVE

Review fire, explosion, burn and fume hazards associated with welding, cutting, grinding, brazing and other spark-producing work, and verify all required controls before starting.

TOPIC EXPLANATION

- Hot work is not limited to activities involving an open flame. Grinding, metal cutting and other heat-producing work can start fires through sparks, slag and hot surfaces.
- Sparks do not remain at the immediate work point. They can travel through platform gaps, penetrations and openings into lower or adjacent areas. Hidden insulation, dust, oil residue or waste can smoulder without being immediately visible.
- Hot work must therefore begin only when the permit is valid, gas testing is acceptable, combustibles are controlled, firefighting equipment is ready and a trained fire watch is present. The fire risk continues after the task has stopped.

REALISTIC SITE SCENARIO

- A maintenance team carried out a short grinding task. Visible combustibles had been removed, but insulation below the platform had not been identified.
- Sparks fell through a platform gap. The fire watch was distracted by another duty and no post-work inspection was completed. The insulation ignited approximately twenty minutes later.
- The event caused production loss, equipment damage and an emergency evacuation. Proper screening, a dedicated fire watch and post-work monitoring would have prevented it.

REMEMBER

Some hot-work fires begin after the task is complete because of hidden embers or heated materials.

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Hazards, controls and daily pre-work verification.

8-10 MINUTES

KEY HAZARDS

- Sparks and slag reaching lower or adjacent areas.
- Ignition of flammable gases, vapours, solvents or dust.
- Residual product or pressure inside lines and equipment.
- Damaged cables, hoses, regulators or connections.
- Welding fumes and inadequate ventilation.
- Changing oxygen or gas levels in enclosed spaces.
- Unsafe handling or storage of gas cylinders.
- Hidden embers and combustion after completion.

CONTROL MEASURES

- Verify the approved hot-work permit.
- Complete and record required atmospheric testing.
- Remove combustibles or protect them with suitable covers.
- Inspect and barricade lower and adjacent areas.
- Assign a trained and dedicated fire watch.
- Provide appropriate extinguishers and fire blankets.
- Inspect cables, hoses, regulators and equipment.
- Provide adequate ventilation and fume control.
- Contain sparks, slag and hot materials.
- Confirm the post-work monitoring period on the permit.

SUPERVISOR TALKING SCRIPT

Team, today's welding, cutting or grinding can turn a small spark into a major fire. We will not start if the permit is not approved, gas testing is unsafe, the fire watch does not understand the role or combustibles remain unprotected. We must inspect lower levels and hidden openings as well. After the work stops, the fire watch will continue for the full period stated on the permit.

DISCUSSION QUESTIONS

- Who is today's dedicated fire watch?
- Where are the nearest suitable extinguisher and alarm point?
- When was gas testing completed and what were the results?
- Can sparks reach lower or adjacent areas?

ATTENDANCE AND APPROVAL

HOT WORK TOOLBOX TALK

Project / Site

Date

Presented by

Work area

No	Full Name	Company / Role	Signature
1			
2			
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Supervisor notes / additional site instructions